PAYPAL AI HACKATHON · "BUILD WHAT'S NEXT WITH PAYPAL AND AI"

PropLock AI

Problem, solution, architecture and workflow, explained for judges, teammates and beginners

Domain
Rentals and real estate, India first

Users
Brokers, agency managers, tenants, owners

Core engine
PayPal Catalog, Orders v2 (AUTHORIZE), Payments, Invoicing

In one sentence: PropLock AI checks a broker's rental ad against facts the broker can't fake, then backs every property visit with a small, refundable PayPal hold. Tenants stop paying for fake listings, and brokers stop driving across town for people who never show up.

1 PropLock in plain words (beginner guide)
2 The problem
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1 PropLock in plain words

If you have never worked with real estate, payments or AI agents, start here. Everything later in the document builds on these few ideas.

The situation today

Imagine you're looking for a flat to rent in Bengaluru. You see an ad: *"Luxury 2BHK, 2 min walk to metro, fully furnished, ₹30,000."* You call the broker and travel across the city. The flat is empty, the metro is a 20-minute walk, and the rent is suddenly ₹40,000. Then the broker asks for ₹10,000 in cash "to hold the flat".

Now look at it from the broker's side. They get dozens of calls a day. They book a visit, drive 45 minutes in traffic, wait at the door, and the tenant never comes. Saying "yes, I'll come on Saturday" costs the tenant nothing, so many people say it to several brokers.

Both sides have good reasons to distrust each other, so everyone protects themselves, and the whole process becomes slow, expensive and full of scams.

What PropLock does, as two simple ideas

Idea 1: a fact-checker for ads

Before an ad goes live, PropLock compares it with things the broker can't change:

- the **owner's agreed rent and deposit** (written down by the agency),
- **real travel times** from a maps service at rush hour,
- the **law** on how big a deposit can be,
- the **owner's own photos** and the GPS location where each photo was taken.

Only ads that pass get a **"Verified"** badge.

Idea 2: a referee who holds a small deposit

To book a visit, the tenant approves a **₹500 hold** on their PayPal account. The money doesn't move yet; PayPal just reserves it. After the visit:

- tenant came but didn't like the flat → hold **cancelled**, no fee;
- tenant wants the flat → the ₹500 becomes the **token** and is subtracted from the first bill;
- tenant never came → the ₹500 **compensates the broker** for the wasted trip.

ANALOGY

Think of a hotel. When you check in, the hotel puts a hold on your card. If nothing goes wrong, the hold disappears and you're never charged. If you damage the room, they charge it. PropLock uses the same PayPal feature, called an **authorization hold**, for property visits. The rules for releasing or keeping the money are enforced by software, using proof such as a door PIN and GPS, instead of one side's word against the other's.

Five words you'll see everywhere

Word	What it means here
Mandate	The owner's written permission for *one* broker to advertise *one* flat, with limits: rent range, maximum deposit, maximum brokerage fee and an end date. No mandate, no listing.
Audit	The automatic check of an ad, stored as a report that shows every comparison made and its result.

Word	What it means here
PASS / FAIL / UNVERIFIABLE	Every check gives one of three answers. *Unverifiable* means "we couldn't check this". It is never quietly turned into a pass; a human manager decides.
Authorization hold	PayPal reserves money on the tenant's account without sending it. Later the hold is either **voided** (released, no fee) or **captured** (actually paid).
AI agent	A program that uses an AI model for one narrow job, like reading an ad and listing its claims, and then hands the result to strict, testable rules. In PropLock, the AI never decides alone whether money moves.

2 The problem

Getting leads is easy for brokers now: portals, WhatsApp groups and chatbots bring plenty. What's broken is the middle of the funnel, turning an inquiry into a real visit and a closed deal, because neither side can trust the other.

2.1 The commitment–trust deadlock

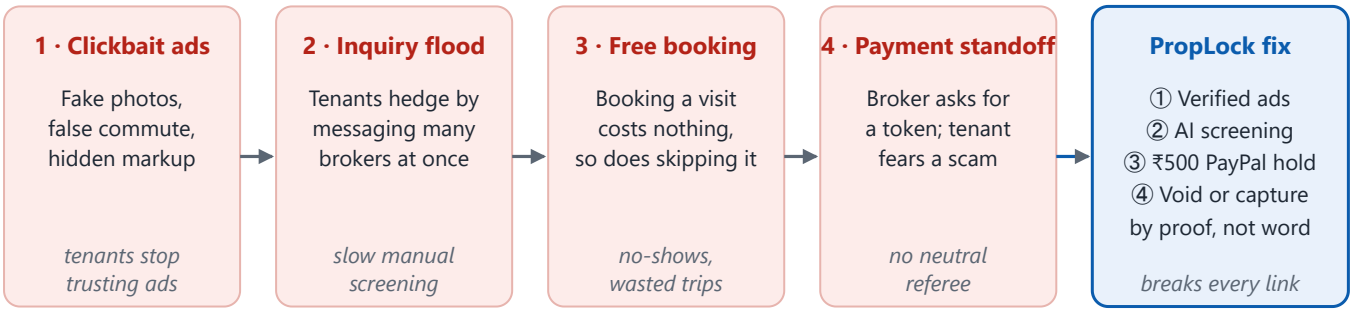
Brokers get no commitment

A broker's stock in trade is their time: driving across town, collecting keys from the owner, giving a tour of up to an hour. A verbal "I'll come Saturday at 4" carries no penalty if the tenant doesn't turn up. The PropLock blueprint estimates that up to 70% of free scheduled visits end in no-shows or last-minute cancellations. Brokers are paid only when a deal closes, so every no-show is unpaid work.

Tenants get no trust

Many listings are clickbait: photos of a nicer flat, a "5 min to metro" that is really a straight-line distance, a rent that goes up after the visit, or a flat that was never available. So tenants won't send token money through a raw payment link to an unknown broker. It looks exactly like an advance-fee scam, and there's no referee to get it back.

2.2 How each failure causes the next



Each failure feeds the next. Fixing only one link (say, a chatbot) leaves the others in place.

Figure 1. The vicious cycle of mutual distrust, and where PropLock steps in.

2.3 The scams people actually run

#	Scam	How it works	Who loses
1	Fake or stolen photos	Photos of a nicer flat copied from another portal or a builder's brochure.	Tenant: wasted visit, broken trust
2	Bait listing	A cheap ad for a flat that doesn't exist or is already rented: "that one's gone, but I have another…"	Tenant; honest brokers lose leads
3	Commute lies	"5 min to metro" measured as a straight line, not the real walk at rush hour.	Tenant: daily commute is far worse
4	Furnishing lies	"Fully furnished" turns out to be a bed and a fan.	Tenant

#	Scam	How it works	Who loses
5	Rent markup	Listed at ₹40k; the owner gets ₹35k and the broker quietly keeps the difference.	Tenant overpays every month; owner is misrepresented
6	Inflated deposit	Asks for 6–10 months of deposit, keeps part of it or passes it on late.	Tenant; it may also break state rules
7	Token-advance scam	"Pay ₹10k to hold the flat", then the broker disappears.	Tenant loses the money
8	Visit / registration fee	Charges a fee before showing any flat.	Tenant
9	Fake broker or owner	Someone with no link to the flat collects advances from several tenants.	Tenants; the real owner's reputation
10	Hidden charges	Extra fees revealed only at signing.	Tenant
11	Double-booking	Takes tokens from several tenants for the same flat.	All but one tenant
—	Tenant no-show	Says "I'll come" to many brokers and turns up for one.	Broker: unpaid time and fuel

2.4 Why the tools that exist today don't fix it

Existing approach	What it does	What's missing
AI lead chatbots	Answer questions on WhatsApp or the web, collect budget, book callbacks.	Booking is still free, so no-shows stay; they don't check whether the ad is true.
AI real-estate agents	Qualify leads, schedule tours, update a CRM, send payment links.	A raw payment link has no refund guarantee, no proof of attendance and no referee.
Portal "verified" badges	Verify the person posting (ID, phone, owner documents).	They verify the *poster*, not each *claim* in each ad.
Photo-compliance tools (e.g. US MLS photo checks)	Detect watermarks, logos, duplicates and staging in photos.	They don't check that the photo shows *this* flat, and they're not tied to any payment.

GAP PROPLOCK FILLS

Each part exists somewhere, but nothing combines **claim-by-claim ad verification** with a **refundable payment hold that releases or pays out based on proof**. That combination is PropLock's core idea.

3 The solution

PropLock AI is a trust and commitment layer between brokers, their agency and tenants. Four AI agents each handle one stage of the deal, and PayPal acts as the neutral referee that holds the money.

CORE IDEA

Money is never sent to the broker at booking time. PayPal **holds** it (an authorization), and the hold is **voided** (released) or **captured** (paid) based on proof: a door PIN, GPS presence and both sides' consent. A small payment becomes a **verifiable signal of intent** that carries no risk for an honest tenant.

3.1 The four agents

FEATURE 1 · TRUST LAYER

Listing Audit & Commute-Truth Agent

Checks every ad before it goes live: photos (A), commute times (B), deposit law (C) and the owner's agreed terms (D). Ads that pass become a signed, verified listing published as a **PayPal Catalog product**.

Being built: phases 0–3 done

FEATURE 2 · SCREENING LAYER

Qualification & Slot-Locking Agent

Chats with the tenant and answers deposit, brokerage and availability questions from the *verified* listing data. It checks budget, move-in date and profile, then puts a 15-minute soft hold on a visit slot, with travel buffers between visits.

Designed

FEATURE 3 · ESCROW ENGINE

PayPal Auth-Escrow & Check-In Referee

Turns the soft hold into a confirmed visit with a ₹500 / $15 **Orders v2 AUTHORIZE** hold. At the door it checks a 4-digit PIN and GPS, then **voids** or **captures** the hold according to fixed rules.

Designed

FEATURE 4 · CLOSING & LEDGER

Token-to-Invoice Ledger & Dispute Copilot

When a tenant locks a flat, it sends a **PayPal Invoice** for deposit and brokerage with the ₹500 already deducted. The broker's dashboard tracks every hold, capture, void and invoice, and answers plain-English questions about them.

Designed

3.2 Every stage of the deal, before and after

Stage	What goes wrong today	How PropLock solves it	Feature
Listing	Fake photos, false commute, hidden markup	Four checks against owner terms, maps, law and photos; only verified ads get published	F1
Inquiry	Window-shoppers look the same as serious tenants	A refundable hold is needed to lock a visit slot	F2, F3
Qualification	Budget and intent checked by hand in chats	The agent checks budget, timeline and profile automatically	F2
Repeat FAQs	Same questions about deposit, brokerage and availability	Answered instantly from the verified listing, not from memory	F1 → F2

Stage	What goes wrong today	How PropLock solves it	Feature
Scheduling	Slot conflicts, no-shows	15-minute soft holds that become hard bookings once the PayPal hold is approved	F2
Commitment	Token collected informally, if at all	`intent=AUTHORIZE` hold, later voided or captured	F3
Trust	Neither side can prove anything	Verified listing + PIN/GPS proof + PayPal as referee	F1, F3
Follow-up	Warm leads go cold	Invoice reminders and slot-expiry alerts	F4
Tracking	Payments scattered across notebooks and screenshots	One ledger of holds, captures, voids and invoices	F4
Compliance	Deposit and refund rules differ by state	Deposit caps checked before publishing, with the rule cited	F1

3.3 Which scam each part stops

Scam (from §2.3)	Stopped by
1 Fake photos	F1 Check A: owner's baseline photos shown publicly, GPS check-in photo, AI same-room comparison
2 Bait listing · 9 Fake broker	Mandates: only a broker the agency assigned to a flat can list it. Logins with roles.
3 Commute lies	F1 Check B: rush-hour route from the owner's verified location. Claims hidden in the ad text are found by AI.
4 Furnishing lies	F1 Check D (owner's registered furnishing) + Check A (photos)
5 Rent markup · 10 Hidden charges	F1 Check D: rent must be inside the owner's range; commission only as a declared, capped brokerage fee
6 Inflated deposit	F1 Check C (state rule) + Check D (owner's maximum)
7 Token scam · 8 Visit fee	F3: the token stays held by PayPal and is released automatically by the rules
11 Double-booking	F2/F3: one active lock per flat, enforced by the system
Tenant no-show	F3: capture after a grace period, only if the broker's GPS proves they were at the flat

4 Solution architecture

Two web apps (the tenant's chat and the broker/manager cockpit) talk to one backend. The backend runs the AI agents, keeps every decision in a database, and is the only part that talks to PayPal.

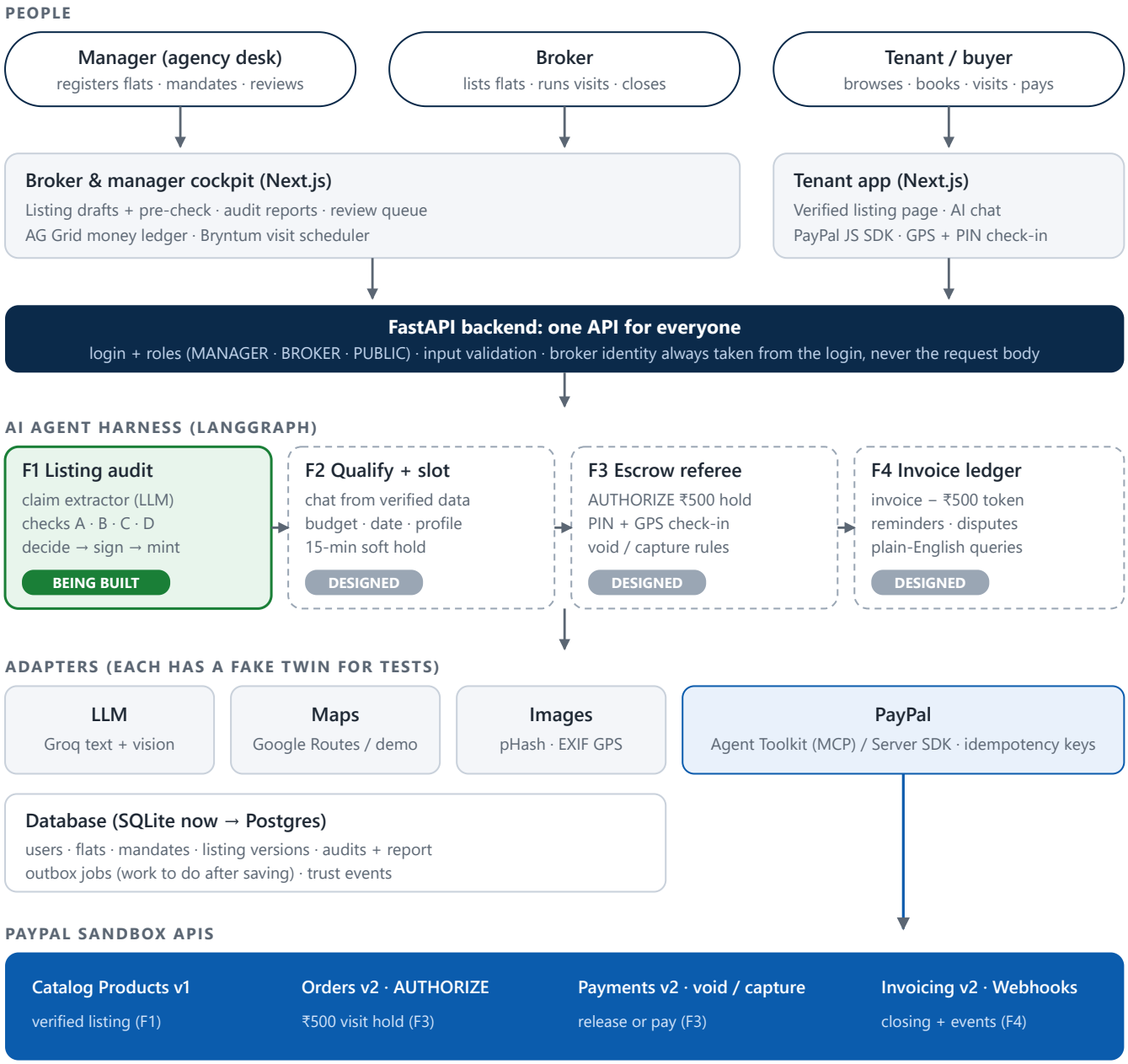


Figure 2. Full-system architecture. Solid green = being built now; dashed = designed, not yet built.

4.1 What each layer does

Layer	Responsibility	Technology
Frontends	The tenant's app shows only verified listings, runs the chat, collects PayPal approval and does the GPS/PIN check-in. The cockpit lets brokers draft and pre-check ads and lets managers review, schedule and reconcile.	Next.js, PayPal JS SDK, AG Grid, Bryntum Scheduler

Layer	Responsibility	Technology
API	One entry point. Checks who is calling and what role they have, validates every input, and returns clear errors (401 not logged in, 403 not allowed, 422 bad data).	FastAPI, Pydantic
Agent harness	Runs each agent as a small graph of steps. Independent checks run in parallel and their results are combined in one decision step.	LangGraph
Domain rules	Plain Python with no network or database: rent-range comparisons, commute tolerance, deposit caps, the final decision. Fast to test and easy to explain.	Python
Adapters	The only code that talks to outside services. Each one has a fake version for tests. A missing key gives “unverifiable”, never a guess.	Groq SDK, httpx, Pillow/ImageHash, PayPal SDK / MCP
Database	The source of truth: who may list what, every version of every ad, every audit report exactly as it was shown, and pending PayPal work.	SQLAlchemy 2, Alembic; SQLite → Postgres
PayPal	Holds and moves the money, publishes verified listings, sends invoices, and reports events back by webhook.	Catalog, Orders v2, Payments v2, Invoicing v2

4.2 Design rules that make it trustworthy

“Unknown” is never “pass”

Every check answers PASS, FAIL or UNVERIFIABLE. A missing API key, a timeout or no map data sends the listing to a human (IN_REVIEW). It never produces a made-up result or a quiet pass.

The AI reads; code judges

The AI model only turns messy text and photos into structured claims, like “2 min walk to Indiranagar Metro”. Whether a claim is true is decided by strict rules. A fooled AI can at worst miss a claim; it can’t award a badge or move money by itself.

Save first, then call PayPal

The signed audit is saved before PayPal is called. A background worker (the “outbox”) then creates the PayPal product, retrying safely with a fixed request ID. A PayPal outage delays the listing; it can’t lose the audit or invent a fake product ID.

Roles and mandates

Only a manager can register flats and give mandates. A broker can list only flats they hold an active mandate for, and their identity comes from their login, so they can’t act as someone else.

IN PLAIN WORDS: WHAT IS AN “API”?

An API is a list of requests one program can send to another, such as “create this listing” or “hold ₹500 on this account”. The PropLock frontends send requests to the PropLock backend, and the backend sends requests to PayPal. PayPal never trusts the frontend directly: only the backend, with its secret keys, can hold, release or capture money.

5 Feature 1 in depth: the trust layer

Feature 1 is the foundation: Features 2–4 answer questions from, take bookings for, and invoice against the verified listing it produces. It's also the part being built right now.

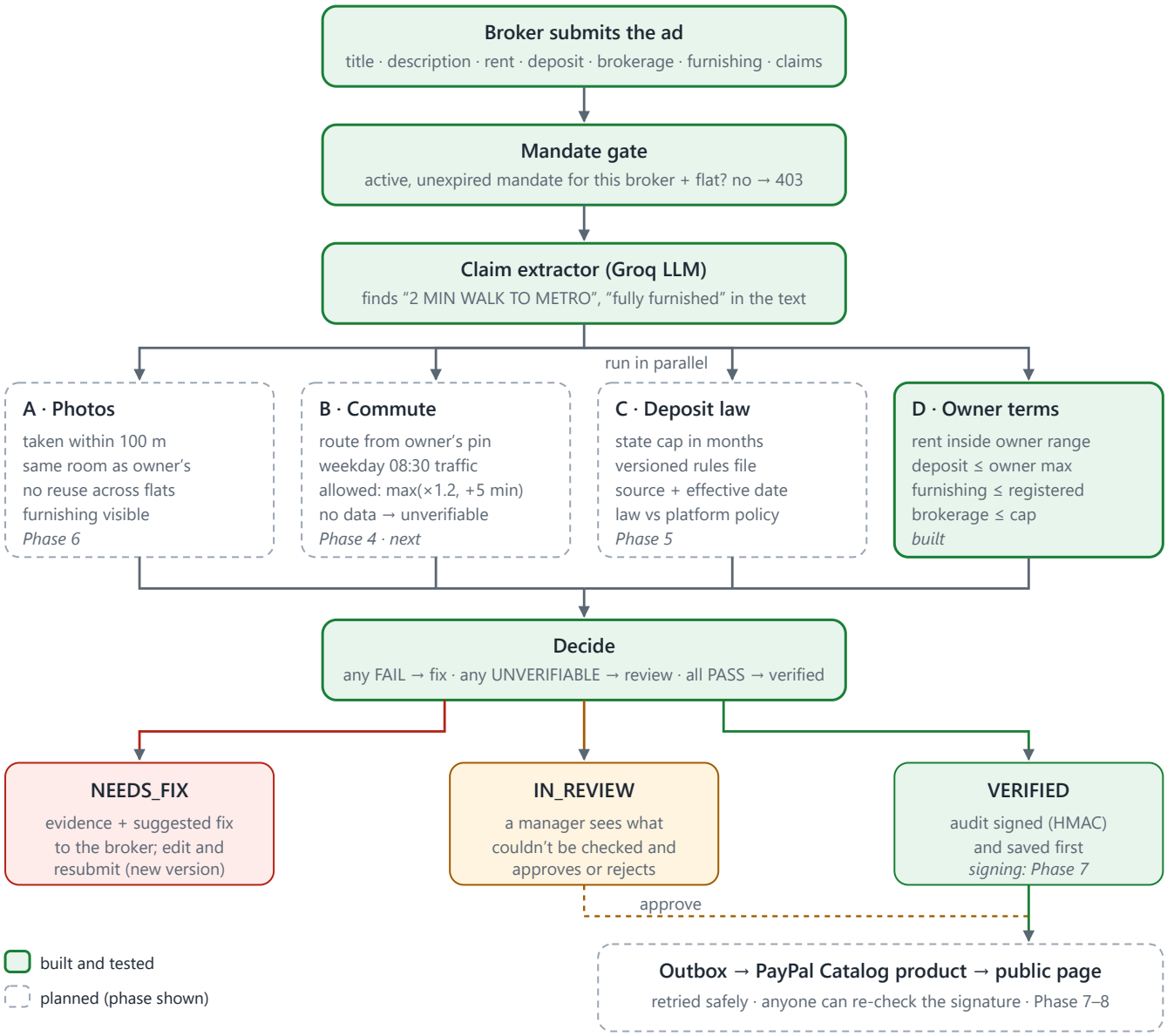


Figure 3. The Feature 1 audit pipeline. Until checks A–C exist, any claim they would judge goes to review instead of passing.

5.1 The four checks

Check	Question	Compared against	Example result
A Photos	Are these real, current photos of *this* flat?	The owner's baseline photos (AI "same room?" comparison that ignores furniture), the photo's GPS (within 100 m of the flat), and our database of photos already used for other flats	FAIL photo taken 3.2 km away

Check	Question	Compared against	Example result
B Commute	Is “5 min walk to metro” true?	Maps route from the owner’s verified location at 08:30 on a weekday. Allowed = max(claimed × 1.2, claimed + 5 min)	**FAIL** claimed 5, actual 22 (allowed 10)
C Deposit law	Is this deposit legal in this state?	A versioned rules file with section, source link and effective date	**FAIL** Chennai: 6 months, cap 3
D Owner terms	Does the ad match what the owner agreed?	The mandate: rent range, maximum deposit, brokerage cap; the flat’s registered furnishing, including furnishing claimed in the ad text	**FAIL** ₹40,000 vs owner range ₹32–35k

5.2 What a stored audit report looks like

Each audit stores the full report exactly as it was shown to the broker. Every comparison is listed, including the ones that passed, and the report names the mandate it was judged against. Below is a real result from the current build, for a dishonest submission on the demo flat in Indiranagar:

Field	Ad says	Owner allows	Result
rent	40000.00	32000.00 – 35000.00	**FAIL**
deposit	140000.00	≤ 70000.00	**FAIL**
furnishing	Fully Furnished	≤ Unfurnished	**FAIL**
brokerage_fee	35000.00	≤ 35000.00 (1 month, counted at most at max rent)	**PASS**

Decision: **NEEDS_FIX**. The broker fixes the ad and resubmits, which creates version 2 with its own audit. Version 1 and its failed audit stay on record as evidence.

5.3 Why it’s built this way

- **Versions, not edits.** Changing a verified ad creates a new version that needs its own audit. Old terms stay on record, which protects tenants in disputes (“the ad said ₹35,000”).
- **One transaction per submit.** The new version, its audit report and the listing’s new status are saved together or not at all, so there’s never a badge without a report.
- **Signed audit (planned).** The verified report is turned into canonical JSON and signed with a server secret (HMAC). Any later change to rent, photos or claims makes the signature fail, and a public `/verify` endpoint lets anyone check it.
- **Commission is a separate fee, not hidden in the rent.** Brokerage is declared in its own field and capped (e.g. one month of rent), so the tenant sees the true move-in cost: rent + deposit + brokerage.

IN PLAIN WORDS: LANGGRAPH

LangGraph lets us draw the agent as a flowchart in code: “first read the claims, then run four checks at the same time, then decide.” Each box is a small function. Because the order is explicit, we can test each box on its own, add a new check by adding one box, and be sure the decision waits until every check has finished.

6 User workflow, end to end

One flat, one broker, one tenant, from the agency signing up the flat to the tenant moving in. Names match the demo data: Rajesh is an honest broker, Priya is a tenant, and the flat is a 2BHK in Indiranagar at ₹35,000 a month.



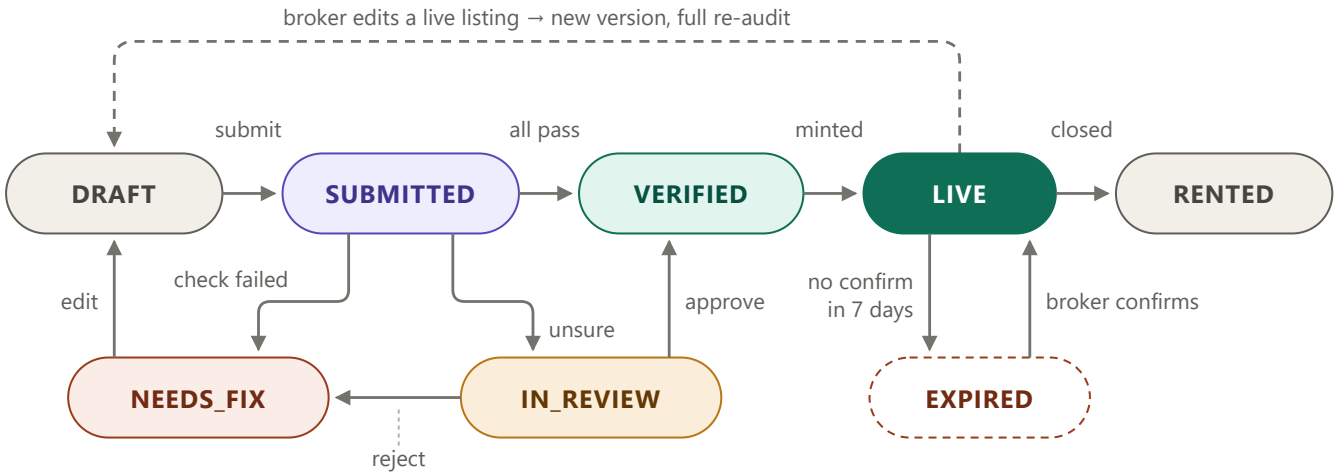
Figure 4. End-to-end workflow across the five parties.

6.1 The four ways a visit can end

The blueprint calls this the anti-cheat matrix. The money only moves when both kinds of proof agree, so neither side can cheat on their own:

What happened	Proof needed	PayPal action	Why nobody can cheat
Tenant visited, doesn't want it	Door PIN entered (or tenant's GPS at the flat)	**void** 100% released, $0 fee · flat stays open	Once the PIN is entered, the no-show penalty can no longer apply. "Pass" frees the slot for the next visitor.
Tenant wants the flat	Both agree: tenant taps Lock, broker approves	**capture** + invoice · flat LOCKED	No capture, no lock. The ₹500 is subtracted from the first invoice, so a serious tenant loses nothing.
Tenant didn't show	Broker's GPS at the flat *and* tenant's GPS absent, after a 30-min grace period	**capture**, which compensates the broker	A broker can't claim a no-show from home; their own GPS must place them at the flat during the slot.
Broker didn't show, or fake flat	Tenant's GPS at the flat + live geo-tagged photo	**void** + listing flagged	The referee releases the hold without the broker's permission, and the listing loses its badge.

6.2 The life of a listing



Solid pills are the happy path. Every edit creates a new signed version, so old terms stay on record for disputes.

Figure 5. Listing states. A failed check sends the ad back to the broker, an unsure one goes to a manager, and a live ad must be re-confirmed every 7 days or it expires.

7 How it fits a broker’s real day

PropLock doesn’t ask brokers to work differently. Every check is attached to something they already do: visit the flat, post the ad, show the flat, close the deal. The goal is less wasted time, not more paperwork.

7.1 How brokers work today

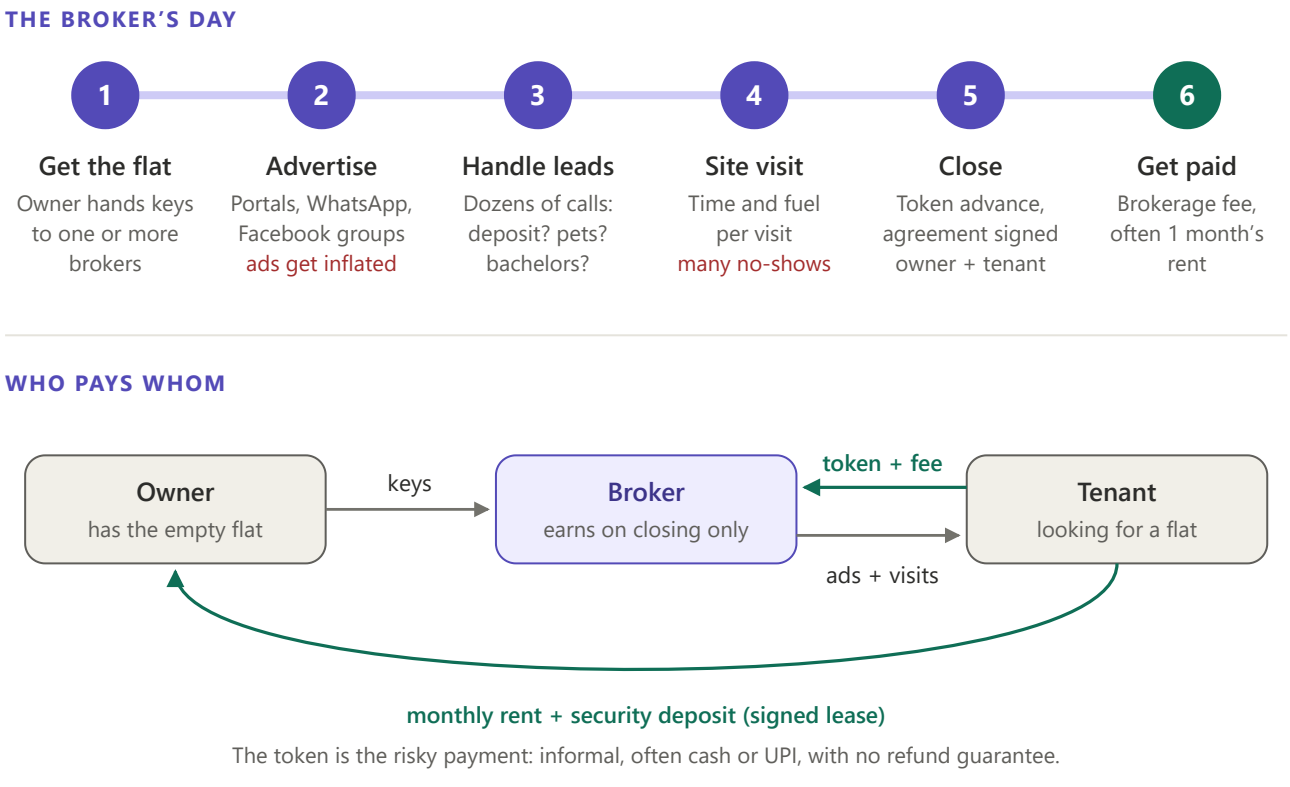


Figure 6. A rental broker’s workflow in Indian metros today. The token is the risky payment.

Based on common practice in Bengaluru, Chennai, Mumbai and Delhi. Fees and customs vary by city. Brokerage is usually one month’s rent (sometimes 15 days), paid separately, usually by the tenant.

7.2 The same day with PropLock

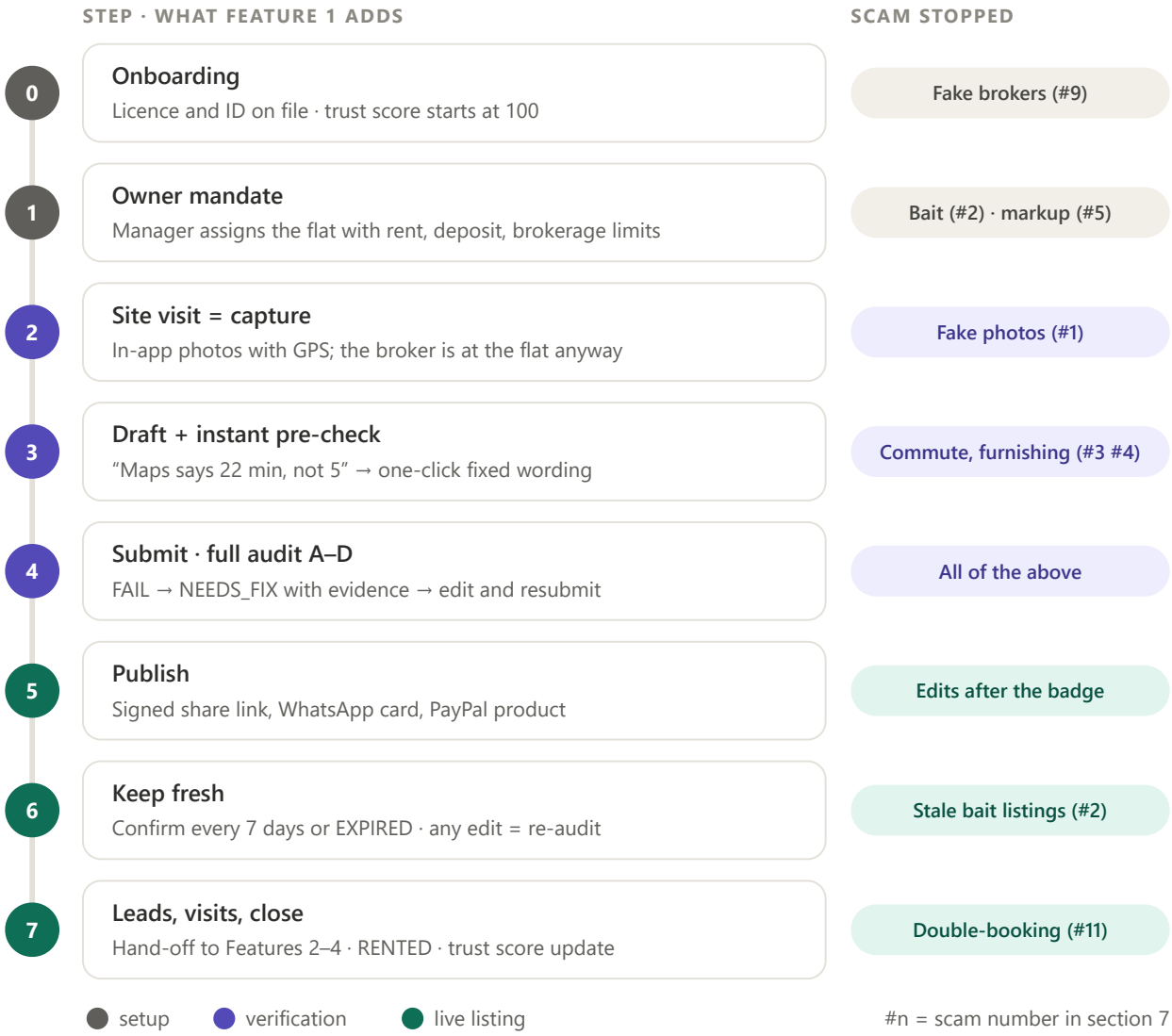


Figure 7. Each step adds one check to something the broker already does. Scam numbers refer to §2.3.

7.3 Step by step: what changes for the broker

Real-world step	Today	With PropLock	Extra effort
Getting the flat	Owner hands over keys, verbally agrees a price	Agency records the owner’s terms; broker gets a **mandate** in the app	None for the broker
Visiting to shoot photos	Takes photos on the phone	Takes a **check-in photo in the app**, which records GPS and time automatically	Seconds; they’re at the flat anyway
Writing the ad	Free text, often exaggerated to beat rival brokers	**Instant pre-check**: “Maps says 22 min, not 5” with one-click corrected wording; nothing saved yet	Fewer rejected ads
Posting	Portals, WhatsApp, Facebook groups	Shares a **verified link / WhatsApp card** with the badge and the owner’s photos	None; one link instead of many uploads

Real-world step	Today	With PropLock	Extra effort
Answering leads	Dozens of calls with the same questions	The AI chat answers from verified data and qualifies the tenant (F2)	Less: only qualified tenants reach the calendar
Site visit	Drives over and hopes they come	The visit is backed by a ₹500 hold; enters the tenant's PIN at the door	One PIN entry; paid if the tenant doesn't show
Closing	Cash/UPI token, agreement, WhatsApp screenshots	Capture + invoice with the token deducted; everything in the ledger (F4)	Less: no manual tracking
Keeping ads fresh	Old ads stay up for months (bait)	Confirm availability every 7 days or the listing expires	One tap a week

WHY AN HONEST BROKER WOULD WANT THIS

- **Paid for no-shows.** A tenant who books and doesn't come forfeits the ₹500, but only if the broker's GPS proves they were there.
- **Stands out from the clickbait.** A verified badge plus a trust score that grows with every first-time-verified listing (+2) and every completed visit (+1).
- **Fewer pointless calls.** FAQs are answered automatically, and only qualified tenants with a hold reach the calendar.
- **Protection in disputes.** Signed listing versions and PIN/GPS logs prove what was promised and who showed up.

COMMISSION VS. RENT MARKUP

Brokers do need to earn. In the normal Indian rental market, commission is a *separate* fee, not added to the rent: the lease is signed by owner and tenant, so a hidden markup comes out at signing. PropLock therefore allows an **asking price anywhere inside the owner's range** and a **declared, capped brokerage fee**, and blocks rent above the owner's maximum. Rent-to-rent and co-living operators who legitimately build margin into the rent can be supported later, with the owner's written consent recorded in the mandate.

8 How it fits a manager's real day

The "manager" is the agency desk, for example "XYZ Properties": the people who sign up flats from owners and hand them to their own or partner brokers. In PropLock they hold the source of truth that every check is measured against.

8.1 What the manager does, and what PropLock does with it

Manager task	Real-world moment	In PropLock
Register the flat	Owner signs up with the agency and hands over keys	Records owner, address, **verified GPS pin**, registered furnishing and baseline photos. Only managers can do this, so nobody can invent an owner.
Agree terms with the owner	"Don't go below ₹32k or above ₹35k; deposit max 2 months; brokerage 1 month"	Written into the **mandate** as min/max rent, maximum deposit, brokerage cap and end date. Check D uses these exact numbers.
Assign brokers	Gives the flat to one or more brokers	One mandate per broker per flat. Revoking it takes effect immediately: the broker can't submit or keep the ad live.
Handle unclear cases	"Is this listing OK?" phone calls	A **review queue** of IN_REVIEW audits showing exactly what couldn't be checked (e.g. no route data). Approve or reject with a note.
Watch broker quality	Reputation by word of mouth	Trust score per broker from trust events: verified first time +2, needs fix −1, photo found online −15, broker no-show −10.
Track money	Notebooks, screenshots, bank statements	One AG Grid ledger of every hold, capture, void and invoice, queried in plain English (F4).
Resolve disputes	One side's word against the other's	Signed listing versions, audit reports, check-in PIN/GPS records and PayPal's own records.

8.2 A manager's week in practice

1. **Monday:** two new owners sign up. The manager registers both flats (address, GPS pin, furnishing, a few baseline photos) and sets the rent range with each owner.
2. **Monday:** gives Rajesh a 60-day mandate for the Indiranagar flat and QuickCash Rentals a mandate for the Chennai flat.
3. **Tuesday:** QuickCash submits the Indiranagar ad at ₹40,000 "fully furnished". It's rejected automatically (NEEDS_FIX) with the evidence. The manager does nothing; the system has already explained why.
4. **Wednesday:** a listing claims "12 min to Manyata Tech Park", a route the demo maps table doesn't cover. It lands in the review queue; the manager checks it and approves with a note.
5. **Saturday:** the ledger shows three visits voided (tenants passed), one captured for a no-show (broker GPS confirmed) and one locked, with an invoice sent with ₹500 deducted.

8.3 Who can do what

Action	Manager	Broker	Public / tenant
Register flats, give or revoke mandates	yes	403	no
Open and submit listings	403	own mandates only	no
Read audit reports	all	own only	verified summary
Approve or reject IN_REVIEW	yes	no	no
Book a visit, approve a hold	—	—	yes

9 Build status, risks and roadmap

The current focus is to get Feature 1 complete and honest, since every other feature depends on it. It is being built in phases, each finished with passing tests and a build-log entry.

9.1 Where Feature 1 stands (4 November 2026)

Phase	Status	What it delivers
0 Foundations	done	FastAPI app, pinned dependencies, settings; Groq proven (text model for claims, vision model that accepts two photos at once)
1 Data, roles, seed	done	11 database tables with migrations; API-key login with MANAGER/BROKER roles; manager endpoints for flats and mandates; demo data
2 First slice (Check D)	done	Three-state results, Check D, decision; LangGraph pipeline; open and submit listing; audits saved in one transaction and read back exactly
3 Claim extraction	done	Groq reads claims hidden in the ad text (tested on English and Hinglish, and against a prompt-injection attempt); furnishing claimed in the text is checked; 78 automated tests
4 Check B commute	next	Tolerance rule, Maps adapter, demo route table; unknown routes go to review
5 Check C compliance	planned	Versioned deposit-rules file with sources and dates
6 Check A photos	planned	GPS check-in photo, duplicate detection, AI same-room comparison
7 Seal and mint	planned	HMAC-signed audit + `/verify`; outbox worker that creates the PayPal Catalog product
8 Broker workflow + demo	planned	Review queue, pre-check, public listing page, availability, trust score, end-to-end demo script

9.2 Changes from the original blueprint

Blueprint said	Now	Why
Gemini 2.5 Flash for vision and text	Groq (`openai/gpt-oss-120b` for text, `qwen/qwen3.8-27b` for vision)	The key that's available. A spike proved both models work, including two images in one request.
Three checks (A–C)	Four checks (A–D) + claim extraction	Without Check D, a broker could charge ₹55k for a ₹35k flat and still be verified. Without extraction, lies in the title were never checked.
VERIFIED or REJECTED	NEEDS_FIX · IN_REVIEW · VERIFIED	Honest brokers get a path to fix their ad, and unclear cases go to a person.
Hash of the listing	HMAC-signed, versioned report covering everything checked	An unsigned hash can be recomputed by anyone, and the old one ignored the broker's photos.

RISKS TO CHECK BEFORE THE DEMO

- **Deadline.** Submissions close **12 November 2026**. Features 2–4 and the frontend are designed but not built, so the demo should lead with Feature 1 end to end plus a narrow slice of the escrow flow (one AUTHORIZE → void / capture in the PayPal sandbox).
- **INR on PayPal.** PayPal stopped domestic payments within India in 2021, so a real ₹500 hold between two Indian accounts may not be possible. Demo in the sandbox in USD ($15) and present INR as the market framing, or name a local partner for production.
- **Authorization timing.** PayPal authorizations have a short honor period (3 days in PayPal's docs) inside a longer validity window (up to 29 days), with re-authorization possible. Visits booked more than a few days ahead need this handled. Check the current docs before relying on the numbers.
- **Legal figures.** The deposit caps in Check C are marked unverified until checked against the official texts. Karnataka's 3–6-month rule is treated as platform policy, not law.
- **Unsourced statistics.** Figures such as "up to 70% no-shows" come from the blueprint and should be presented as estimates unless a source is added.

9.3 How it maps to the hackathon

The hackathon has no fixed tracks; its one rule is that a project must meaningfully use both PayPal and AI. PropLock does both at its core: AI agents decide *when* money should move, and PayPal holds, releases, captures and invoices it. Relevant prize categories include **Best Use of Agentic Commerce**, **Best Use of PayPal + AI** and **Most Impactful**, plus the sponsor prizes for AG Grid (ledger), Bryntum (visit scheduler), APIMatic (type-safe PayPal SDK use) and Render (hosting).

10 Glossary

Term	Meaning
Agentic commerce	Software agents that don't just chat but take real commercial actions (hold, capture, void, invoice) within strict rules.
Audit / audit report	The stored result of checking one version of an ad: every comparison, its verdict and its source.
Authorization (PayPal)	Money reserved on the buyer's account but not transferred. Created with Orders v2 `intent=AUTHORIZE`.
Capture	Turning an authorization into a real payment.
Void	Cancelling an authorization; the money is released with no fee.
Catalog product (PayPal)	A product record in PayPal. PropLock publishes each verified listing as one, linked to its signed audit.
Invoice (PayPal)	A formal bill sent through PayPal; here deposit + brokerage minus the ₹500 token.
Mandate	The agency's permission for one broker to list one flat, with rent, deposit and brokerage limits and an end date.
Ground truth	Facts the broker can't change: the owner's terms, the flat's GPS pin, the registered furnishing.
UNVERIFIABLE	"We couldn't check this." Always sent to a person; never treated as a pass.
NEEDS_FIX / IN_REVIEW / VERIFIED	Outcome of an audit: a claim is wrong / something couldn't be checked / everything passed.
LLM	Large language model, the AI that reads text (here via Groq). Used only to *extract* claims.
Prompt injection	Text inside an ad that tries to give the AI instructions ("ignore your rules"). PropLock treats the ad as data and judges claims with code.
LangGraph	A library for building AI agents as explicit graphs of steps, with parallel branches and a final join.
MCP / Agent Toolkit	Model Context Protocol: a standard way for AI agents to call tools. PayPal's Agent Toolkit exposes PayPal actions to agents this way.
HMAC signature	A code computed from the report and a server secret. If any byte of the report changes, the code no longer matches.
Outbox	A table of "work to do after saving", such as creating the PayPal product. A worker retries it until it succeeds.
Idempotency key	A fixed ID sent with a request so that retrying it never creates a duplicate (e.g. two PayPal products).
pHash	A short fingerprint of an image that stays similar when the image is resized or recompressed; used to spot reused photos.
Geofence	A circle around the flat's GPS pin; being inside it during the slot counts as being present.

Sources: PropLock AI Product Blueprint; PropLock Feature 1 Technical Review (22 Oct 2026); Feature 1 rebuild plan and build log (repository `docs/`); PayPal AI Hackathon announcement on the PayPal Developer Community blog (30 Sep 2026).